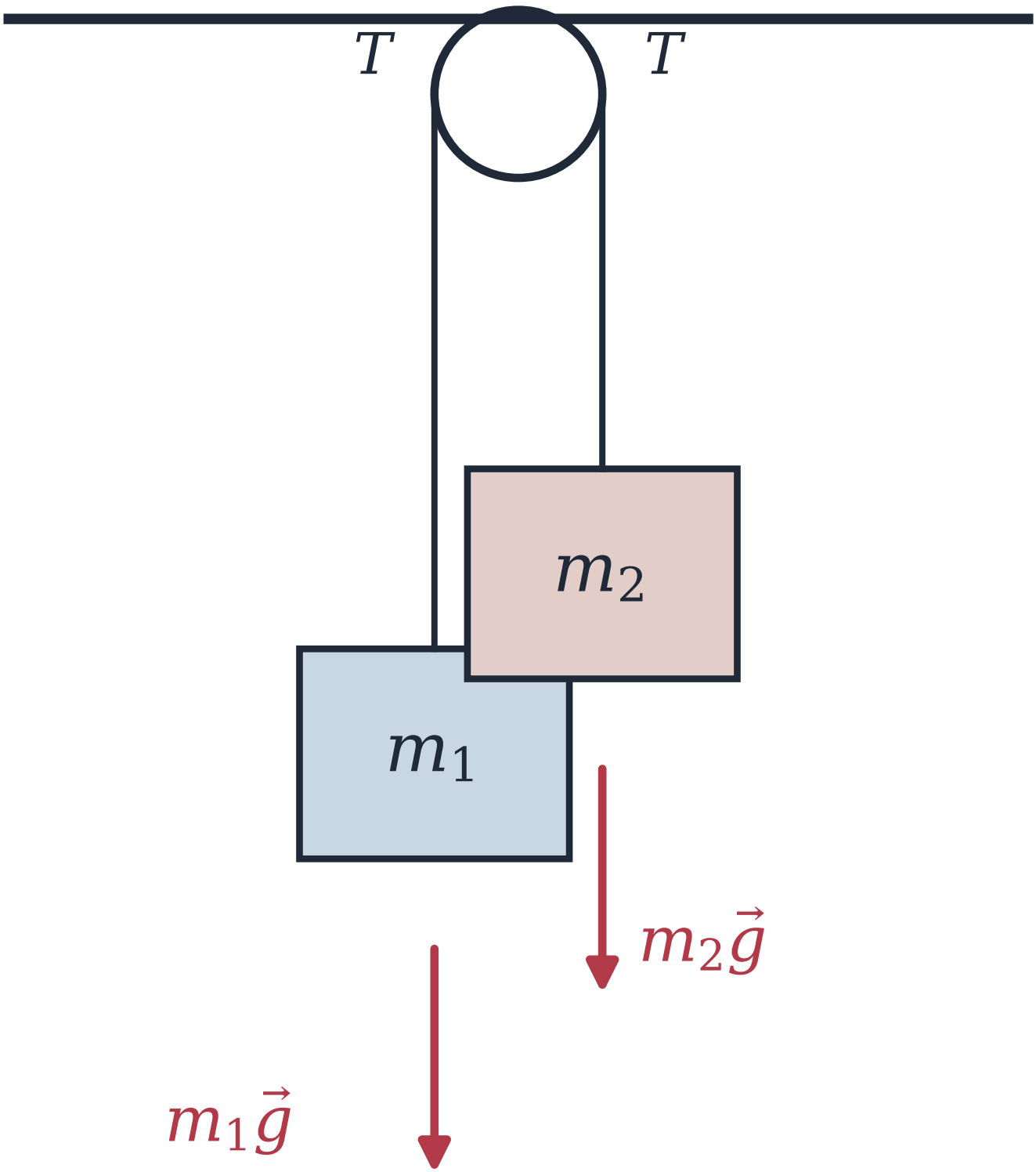

$$a = \frac{(m_1 - m_2)g}{m_1 + m_2}$$